

Ibrahim Khalil Andoulsi

AI & Software Engineer

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EXPERIENCE

• PwC

Feb 2026 - Jul 2026

AI Engineering Intern

- Designed and developed an **Agentic AI** assistant embedded in Power BI, enabling directors to analyze reports, interpret KPIs, apply filters, navigate pages, and create custom visuals via natural-language and voice commands.
- Built a modular architecture combining an **Angular SPA** integrating Power BI Embedded and a FastAPI backend orchestrating an 11-node **LangGraph** agentic workflow for routing, memory, and automated tool execution.
- Engineered an intelligent report pipeline leveraging Power BI APIs and PBIX semantic-model parsing to ground **Azure OpenAI** LLM contexts directly in active visuals, active filters, measures, and complex DAX expressions.
- Integrated Microsoft **Entra ID** and MSAL to secure real-time execution while strictly enforcing enterprise RLS.
- Implemented and validated automated dashboard interactions through Power BI JavaScript APIs, achieving 100% accuracy on basic actions such as page navigation and targeted filtering, 90% accuracy on on-demand visual creation with filtered data, and 75% accuracy on analytical question answering and KPI interpretation.

• Bi'nergy

Jul 2025 - Aug 2025

AI Engineering & MLOps Intern

- Engineered an end-to-end anomaly detection pipeline for industrial sensor time-series data collected per feature at minute-level resolution over approximately one year, with **MLflow** used to track experiments and model runs.
- Built **ETL** and feature-engineering workflows to clean and transform raw machine data, improving model performance from 60% to 92% and ultimately achieving a 96.2% F1-score with 96.9% fault recall using XGBoost.
- Deployed it as a **Dockerized** application with a **CI/CD** pipeline, and implemented **Prometheus/Grafana**-based observability to monitor latency, throughput, error rates and system resource usage, with automated email alerting.

• OSS

Jul 2024 - Aug 2024

AI Software Engineering Intern

- Engineered and deployed an internal **Full-Stack** asset and maintenance management platform using the **MERN** stack, with SQL managing structured inventory data and MongoDB handling user authentication.
- Implemented a maintenance ticket triage workflow combining text features from issue descriptions with structured numerical inputs to prioritize and route requests using a Random Forest classifier.

EDUCATION

• Engineering Degree in Advanced Technologies

2023 – 2026

Machine Learning, Deep Learning, Cloud Infrastructure, Big Data, IoT, Computer Vision

ENSTAB

• Preparatory Cycle in Mathematics and Physics

2021 – 2023

Strong foundation in Mathematics, Linear Algebra, Calculus, and Analytical Problem-Solving

IPEIN

PROJECTS

• DocMind

Personal Project

Multimodal RAG assistant with grounded citations

- Built a document assistant over **Qdrant** with hybrid retrieval (dense + BM25) and reranking across PDF/-DOCX/PPTX/XLSX, raising top-3 retrieval accuracy from 58% to 92.1% on a 100-question Ragas eval set.
- Extended grounding to charts and images via a local **VLM**, enabling page-level cited answers.

• NutriGen

Personal Project

Governed LLM recipe generation platform with full MLOps pipeline

- Fine-tuned **SmolLM2** with **QLoRA** on RecipeNLG (2M+ recipes) with an **AI governance** layer enforcing nutritional constraints (calories, macros, allergens) and full audit logging, achieving 89% constraint satisfaction rate.
- Deployed microservices on **Kubernetes** with HPA; decoupled inference via **Kafka**; automated CI/CD with **Jenkins**.

HACKATHONS & AWARDS

• Green AI Hackathon – 1st Place Winner

ENSTAB

- Designed and developed an agentic AI system to reduce household energy consumption.

Dec 2025

• PwC Internal AI Hackathon – 4th Place

PwC

- Developed an AI-driven solution in PwC's internal hackathon to improve productivity.

May 2026

TECHNICAL SKILLS

Programming Languages: Python, C, C++, JavaScript, TypeScript, SQL, Html, CSS

Dev Frameworks: FastAPI, Angular, ReactJS, NextJS

AI & ML: Agentic AI, RAG, LLM Fine-tuning, LangGraph, HuggingFace, PyTorch, TensorFlow, Scikit-learn, Ragas

Cloud, DevOps & MLOps: Azure, Docker, Kubernetes, GitHub Actions, Jenkins, Kafka, MLflow, DVC, Prometheus, Grafana, Git, Github Actions, GitLab

Databases & Data Infrastructure: Qdrant, MongoDB, PostgreSQL / SQL Server, Power BI

LANGUAGES

English · B2 (TOEIC)

French · Fluent

Arabic · Native